Data Management Project

Project Report

Submitted to,

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FIRST YEAR

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Revision History

S#	Version	Contributors	Date	Description of change
1	Version 1	Fayyaz Hussain Shah	30/11/2025	Document created
2	Version 2	Fayyaz Hussain Shah	1/12/2025	Document Revised
3	Version 3	Fayyaz Hussian Shah	16/12/2025	Document Revised
4	Version 4	Fayyaz Hussian Shah	24/01/2025	Document Revised

1 Introduction

1.1 Problem Background

Understanding the well-being of nations is becoming increasingly important for policy making, global development, and social research. The World Happiness Report provides annual scores for countries based on income, health, freedom, social support and other factors.

However, the publicly available World Happiness dataset is often limited to specific years (e.g., 2006–2020 in Kaggle). To perform long-term trend analysis, it becomes necessary to collect additional years from an external API, merge them, assess data quality, and store them properly.

This project builds a full ETL (Extract–Transform–Load) pipeline for the World Happiness dataset, integrating CSV and API sources into a unified analysis-ready dataset.

1.2 Project Goals

- Collect historical data from a CSV dataset (2006–2020)
- Collect recent data from an API source (2021–2024)
- Integrate both datasets into a single standardized dataset
- Assess and improve data quality
- Store the final dataset in SQL and NoSQL databases
- Perform Exploratory Data Analysis (EDA)
- Identify trends in Life Ladder and Log GDP per capita

1.3 Research Questions

- What is the relationship between income (Log GDP) and happiness (Life Ladder)?
- Top Performers of Life Ladder and GDP
- Evolution of GDP and Life Ladder Globally
- Evolution of Italy's Life Ladder and GDP throughout.

1.4 Technical Brief

1.4.1 Data Sources

The Data source 1 (Kaggle Dataset) contains the following data for year 2005-20,

- Country Name
- Year

- Log (GDP per capita)
- Happiness Score

This Data set has some missing values of GDP for some countries for some years and also missing some data of entire year for some countries, which will be handled in Data Transofmraiton section ahead.

The Data source 2 (World Bank open API) contains the following data for year 2001-24,

- Country Name
- Year
- GDP per capita
- Happiness Score

The API returns the data in JSON format, and the GDP is not logged along with few column names mismatching with the source 1, all of which along with the merging the data from these sources into one Data frame and SQL DB, will be handled in Data Transformation section ahead

1.4.2 GDP and Happiness Score

What is GDP and how it is calculated? Total monetary value of all final goods and services produced within a country in a period (usually a year). **GDP=C+I+G+(X−M)** **C**– Consumption - Household spending on goods/services (food, rent, transport) **I** – Investment - Business spending on machinery, buildings, inventory; residential construction **G** – Government Spending - Public services, infrastructure, salaries (excluding transfers) **X – M** (Net Exports) - Exports minus imports; measures international trade GDP per capita: **GDP/Population**	**What is Life Ladder Score and how it is calculated?** Subjective well-being measure: people rate their life from 0 (worst) to 10 (best). Survey method: “Imagine a ladder from 0–10. On which step do you feel you stand today?” Country score: Average of all survey responses. Other Factors affecting Life Ladder: • Social support • Healthy life expectancy • Freedom to make life choices • Generosity • Perceptions of corruption

Log GDP per Capita : **ln(GDP/Population)**	

2 Data Acquisition

2.1 Source 1: Kaggle Dataset	2.2 Source 2: Open API World Bank
https://www.kaggle.com/datasets/ajaypalsinghlo/world-happiness-report-2021	**http://api.worldbank.org/v2/country/all/indicator/NY.GDP.PCAP.CD?date=2020:2025&format=json&per_page=20000**
`df = pd.read_csv('world-happiness-report.csv')`	

```python
# Years 2020-2025
start_year = 2020
end_year = 2025

# API URL for GDP per capita for all countries
api_url = (f"http://api.worldbank.org/v2/country/all/indicator/NY.GDP"
           f".PCAP.CD?date={start_year}:{end_year}&format=json&per_page=20000")

response = requests.get(api_url)
data = response.json()

# Extract relevant data
records = []
for entry in data[1]:  # data[0] has metadata
    country = entry['country']['value']
    year = int(entry['date'])
    gdp_per_capita = entry['value']  # Can be None
    records.append({
        'Country name': country,
        'year': year,
        'GDP per capita': gdp_per_capita
    })

df_api = pd.DataFrame(records)
```

3 Data Quality Assessment & Enrichment -Pre Integration

Source 1 Kaggle Dataset	Source 2 API Data
3.1 Completeness - Handling Missing Values	

Filling Null Values for GDP with Median – since it's robust to Outliers
 	
`df.isnull().sum()`
`df['Log GDP per capita'] = df['Log GDP per capita'].fillna(df['Log GDP per capita'].median())`
	Filling Null Values for GDP with Median – since it's robust to Outliers
	
	`df_api ['GDP per capita'] = df_api ['GDP per capita'].fillna(df_api ['GDP per capita'].median())`

Soucrce 1 had some countries missing data for few years hence, they were filled by interpolation
 	
`def fill_missing_years(df):`
`  # Find the range of years`
`  df['year'] = df['year'].astype(int)`
 	
`  # Get full range of years in dataset`
`  year_min = df['year'].min()`
`  year_max = df['year'].max()`
 	
`  # Get list of countries`
`  countries = df['Country name'].unique()`
 	
`  # Create full grid`
`  full_grid = pd.MultiIndex.from_product(`
`  [countries, range(year_min, year_max + 1)],`
`  names=['Country name', 'year']`
`  ).to_frame(index=False)`
 	
`  # Left merge ensures all countries x years exist`
`  df_full = full_grid.merge(df, on=['Country name', 'year'], how='left')`
 	
`  # Sort by country and year`
`  df_full = df_full.sort_values(['Country name', 'year'])`
 	
`  # Numeric columns to fill`
`  numeric_cols = ['Life Ladder', 'Log GDP per capita', 'Social support',`
`  'Healthy life expectancy at birth', 'Freedom to make life choices',`
`  'Generosity', 'Perceptions of corruption']`
 	
`  # Interpolate missing values`
`  df_full[numeric_cols] = df_full.groupby('Country name')[numeric_cols].transform(lambda g: g.interpolate())`
 	
`  # Forward/backward fill if first/last years were missing`
`  df_full[numeric_cols] = df_full.groupby('Country name')[numeric_cols].transform(lambda g: g.ffill().bfill())`
`  return df_full`	

3.2 Consistency - Handling Duplication & Formatting

```
df = df.drop_duplicates()

df['Country name'] = df['Country name'].str.strip()

df['Country name'] = df['Country name'].str.title()

df["year"] = df["year"].astype(int)
[25]
```

Keeping only the needed columns, discarding the rest

```
df = df[['Country name', 'year', 'Life Ladder', 'Log GDP per capita']]
```

Source 2 had some entries as grouped countries and their relevant data, hence it needed to be removed

```
df_api = df_api[df_api["Country name"].isin(df["Country name"])]
```

Calculating the Log GDP for it to be consistent with Source 1, and keeping only the needed columns, discarding the rest

```
# Calculate Log GDP per capita
df_api['Log GDP per capita'] = np.log(df_api['GDP per capita'])

# Drop original GDP per capita if not needed
df_api = df_api[['Country name', 'year', 'Log GDP per capita']]
```

4 Data Integration

```
# Integration
df_merged = pd.concat([df, df_api], ignore_index=True)

# Sorting
df_merged = df_merged.sort_values(['Country name', 'year']).reset_index(drop=True)
```

5 Data Quality Assessment & Enrichment -Post Integration

Since the data from API did not contain the happiness score, it had to be populated via Interpolation
Note: The step of filling null values via median is unnecessary since there will be no numm values left after interpolation

```
df_merged.isnull().sum()
df_merged['Life Ladder'] = df_merged.groupby('Country name')['Life Ladder'].transform(lambda x: x.interpolate())
df_merged['Life Ladder'] = df_merged['Life Ladder'].fillna(df['Life Ladder'].median())
```

```
df_merged.isna().mean()*100
[26]
Country name          0.0
year                  0.0
Life Ladder           0.0
Log GDP per capita    0.0
dtype: float64
```

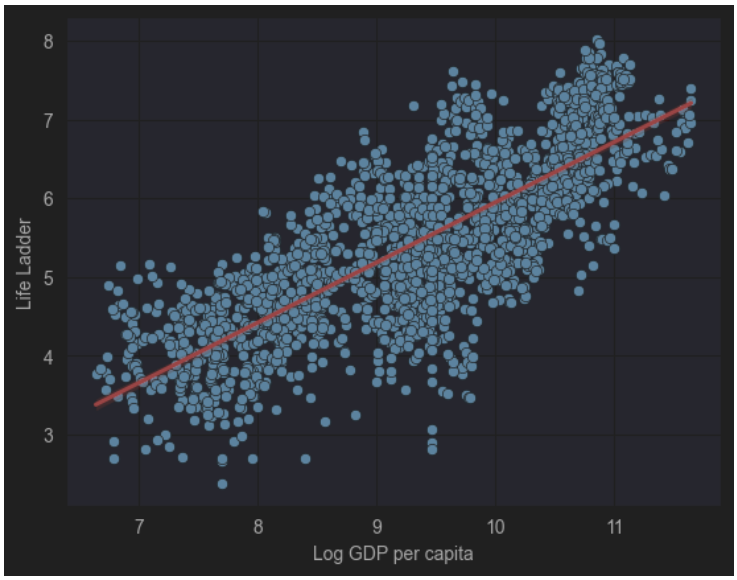
6 Database Storage - SQL

```
conn = sqlite3.connect("WHR.db")
df_merged.to_sql("World_Happiness_Report", conn, if_exists="replace", index=False)
conn.close()
```

7 Exploratory Data Analysis

7.1 Relationship of the GDP of a country and the happiness of its people

```
sns.scatterplot(x='Log GDP per capita', y='Life Ladder', data=df_full)
sns.regplot(x='Log GDP per capita', y='Life Ladder', data=df, scatter=False, color='red')
plt.show()
```



Strength of relationship

The closeness of the points to the regression line shows the strength and the opposite reflects weakness.

Direction of relationship

The upward slope shows partial direct proportionality – partial because of the not so strong relationship

Exceptions – Outliers

Countries above the red line: happier than expected for their GDP

Countries below the line: less happy than expected for their GDP

Summary

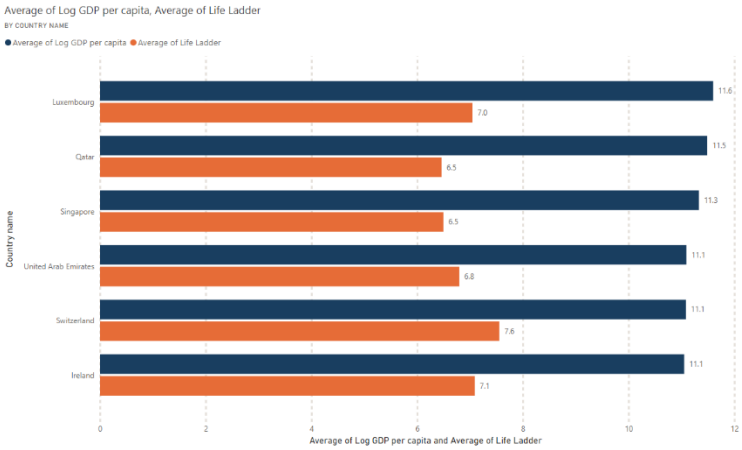
Scatter + regression line shows how Life Ladder changes with Log GDP per capita.

Each point represents a country-year, and the red line shows the average trend.

Points close to the line mean strong relationship; outliers show exceptions.

7.2 Top Performers of GDP and Life Ladder

These visual studies the top performing countries throughout the years and see if they are commonly performing for both or not.



The top performing countries do show a strong positive connection between the GDP and Happiness score however within them there are many contradictory outliers

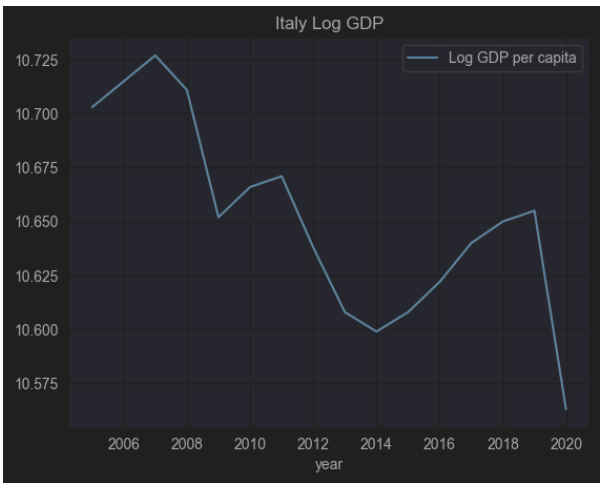
There are various Middle East countries who couldn't achieve the same happiness score even after matching the GDP with the top runners

This concludes that though Happiness score has strong ties with the GDP but it is not the only contributing factor

Visual 2

NOTE: The Graph on top is incomplete. Please click on the link above to view the complete

7.3 Evolution of Life Ladder and GDP in Italy thought out the time

```
country = "Italy"
df[df["Country name"] == country].plot(x="year", y="Life Ladder", title=f"{country} Life Ladder Trend")
plt.show()
✓ [21] 193ms
```

```
country = "Italy"
df[df["Country name"] == country].plot(x="year", y="Log GDP per capita", title=f"{country} Life Ladder Trend")
plt.show()
✓ [22] 192ms
```